Artificial Intelligence and Machine Learning Notes

Introduction to Artificial Intelligence

Artificial Intelligence (AI) is a branch of computer science that focuses on creating intelligent systems capable of performing tasks that normally require human intelligence. These tasks include learning, reasoning, problem-solving, decision-making, natural language understanding, image recognition, speech recognition, and planning. AI is widely used in healthcare, education, finance, transportation, cybersecurity, manufacturing, and entertainment.

AI systems improve their performance by learning from data and identifying patterns. Modern AI relies heavily on machine learning and deep learning algorithms to solve complex real-world problems.

Types of Artificial Intelligence

Artificial Intelligence can be divided into three categories.

Narrow AI

Narrow AI is designed to perform a single specific task. Examples include recommendation systems, spam filters, voice assistants, and facial recognition systems.

General AI

General AI refers to machines capable of performing any intellectual task that a human can perform. This type of AI is still under research.

Super AI

Super AI represents machines that exceed human intelligence in every aspect. It is a theoretical concept and has not yet been achieved.

Machine Learning

Machine Learning is a subset of Artificial Intelligence that enables computers to learn from data without being explicitly programmed.

The machine learning workflow generally consists of:

- Data Collection
- Data Cleaning
- Feature Engineering

- Model Training
- Model Evaluation
- Model Deployment

Machine learning improves accuracy as more relevant data becomes available.

Types of Machine Learning

Supervised Learning

Supervised learning uses labeled data to train models.

Examples include:

- House Price Prediction
- Email Spam Detection
- Student Grade Prediction
- Disease Diagnosis

Popular algorithms include:

- Linear Regression
- Logistic Regression
- Decision Tree
- Random Forest
- Support Vector Machine

Unsupervised Learning

Unsupervised learning identifies hidden patterns without labeled data.

Examples include:

- Customer Segmentation
- Market Basket Analysis
- Topic Modeling

Popular algorithms:

- K-Means Clustering

- DBSCAN
- Hierarchical Clustering
- PCA

Reinforcement Learning

Reinforcement learning allows an agent to learn by interacting with an environment.

Applications include:

- Robotics
- Self-driving Cars
- Game Playing
- Resource Optimization

The agent receives rewards for good actions and penalties for incorrect actions.

Deep Learning

Deep Learning is a specialized area of Machine Learning based on Artificial Neural Networks.

It performs well in:

- Image Classification
- Speech Recognition
- Language Translation
- Object Detection
- Medical Imaging

Popular deep learning frameworks include TensorFlow and PyTorch.

Natural Language Processing

Natural Language Processing (NLP) enables computers to understand, interpret, and generate human language.

Applications include:

- Chatbots

- Language Translation
- Sentiment Analysis
- Text Summarization
- Question Answering
- Document Classification

Large Language Models such as GPT, Llama, Qwen, and Gemini are examples of NLP systems.

Large Language Models

Large Language Models are trained using billions of words collected from books, websites, articles, and other sources.

They can perform tasks such as:

- Writing code
- Answering questions
- Summarizing documents
- Translating languages
- Generating reports
- Creating emails
- Solving programming problems

Popular models include:

- GPT-4o
- GPT-4o mini
- Llama 3.2
- Qwen 2.5
- Claude Sonnet
- Gemini Flash

Ollama

Ollama is a tool that allows users to run Large Language Models locally on their computer.

Advantages include:

- Offline execution
- Better privacy
- No API cost
- Faster experimentation
- Local model management

Common commands include:

- ollama pull
- ollama run
- ollama list
- ollama ps

LangChain

LangChain is a framework used to build applications powered by Large Language Models.

Key components include:

- Prompt Templates
- Chains
- Memory
- Tools
- Agents
- Output Parsers
- Retrievers
- Vector Stores

LangChain simplifies the development of chatbots, question-answering systems, and Retrieval-Augmented Generation applications.

ChromaDB

ChromaDB is an open-source vector database designed for storing embeddings.

Main features include:

- Fast similarity search
- Metadata filtering
- Persistent storage
- Easy integration with LangChain
- Semantic search support

Typical workflow:

1. Load documents
2. Split into chunks
3. Generate embeddings
4. Store embeddings
5. Perform similarity search
6. Retrieve relevant context

Embeddings

Embeddings are dense numerical vectors representing the meaning of text.

Advantages:

- Capture semantic relationships
- Improve search quality
- Support recommendation systems
- Enable Retrieval-Augmented Generation

Popular embedding models include:

- all-MiniLM-L6-v2
- BAAI bge-small
- OpenAI text-embedding models

Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) combines document retrieval with a Large Language Model.

The workflow is:

1. User asks a question.
2. Question is converted into an embedding.
3. Similar document chunks are retrieved.
4. Retrieved context is added to the prompt.
5. The LLM generates an answer using the retrieved information.

Benefits include:

- Improved factual accuracy
- Reduced hallucinations
- Ability to answer domain-specific questions
- Easy updating of knowledge without retraining the model

Vector Databases

Vector databases store embeddings instead of traditional rows and columns.

Popular vector databases include:

- ChromaDB
- FAISS
- Pinecone
- Weaviate
- Milvus

They are commonly used in semantic search, recommendation systems, document retrieval, and AI assistants.

Applications of AI

Artificial Intelligence is used in many domains.

Healthcare:

- Disease prediction
- Medical imaging
- Drug discovery

Finance:

- Fraud detection
- Credit scoring
- Risk analysis

Education:

- Intelligent tutoring
- Personalized learning
- Automatic grading

Transportation:

- Route optimization
- Autonomous vehicles
- Traffic prediction

Business:

- Customer support
- Sales forecasting
- Inventory management

Cybersecurity:

- Threat detection
- Malware analysis
- Network monitoring

Future of AI

Artificial Intelligence continues to evolve rapidly. Modern systems integrate Large Language Models, Retrieval-Augmented Generation, autonomous agents, and multimodal capabilities. Responsible AI, transparency, fairness, and privacy are becoming increasingly important as AI systems are adopted across industries.